

STUDENT PERFORMANCE ANALYTICS SYSTEM

1. Project Title

Student Performance Analytics System

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2. Objective

The objective of this project is to build a Python-based Student Performance Analytics System that reads student data, processes and analyzes it using NumPy and Pandas, and produces meaningful results such as marks summary, averages, grades, pass/fail statistics, subject-wise performance, and top performers.

3. Technologies Used

- Python
- NumPy
- Pandas
- CSV file for storing student data

4. Dataset Description

The dataset is stored in a CSV file named students.csv. It contains 15 student records.

The important columns are:

- Student ID
- Name
- Department
- Python
- DBMS
- DSA
- OS
- JavaScript
- Attendance

The marks are used to calculate total marks, average marks, grades, and performance. Attendance is displayed along with the student's performance.

5. Implementation

1. The student data is stored in a CSV file.

2. Pandas is used to read the CSV file and organize the data in a DataFrame.
3. NumPy is used for numerical calculations such as sum, mean, maximum and minimum.
4. A function is created to assign grades based on the student's average.
5. A loop is used to process each student.
6. Total marks and average marks are calculated for each student.
7. Students are classified as Pass or Fail using a defined condition.
8. Subject-wise average marks are calculated.
9. Class average, highest average, lowest average and best subject are identified.
10. The top-performing student is displayed in the final output.

6. Key Features

- Student data loading using Pandas
- Total marks calculation
- Average marks calculation
- Grade calculation
- Pass/Fail identification
- Attendance display
- Subject-wise performance analysis
- Highest and lowest performance
- Overall class average
- Top-performing student identification

7. Output / Screenshots

EXPLORER

PROJECT

.venv

.vscode

main.py

Students.csv

main.py

Students.csv

Students.csv

1 Student_ID,Name,Department,Python,DBMS,DSA,OS,JavaScript,Attendance

2 STD001,Aarav Sharma,AIML,85,78,92,80,91,92%

3 STD002,Ananya Iyer,AIML,94,88,85,65,74,96%

4 STD003,Rohan Verma,AIML,62,58,70,80,91,90%

5 STD004,Sneha Patel,AIML,78,82,75,72,55,74%

6 STD005,Vikram Rao,AIML,88,91,84,86,89,88%

7 STD006,Priya Nair,AIML,71,75,88,82,80,95%

8 STD007,Rahul Gupta,AIML,55,64,60,58,70,81%

9 STD008,Pooja Kulkarni,AIML,90,85,94,90,92,92%

10 STD009,Aditya Singh,AIML,82,79,78,75,87,94%

11 STD010,Kavya Joshi,AIML,67,72,81,79,84,89%

12 STD011,Manish Mehta,AIML,76,70,68,66,73,85%

13 STD012,Divya Menon,AIML,95,92,89,91,96,98%

14 STD013,Arjun Reddy,AIML,58,62,72,70,60,76%

15 STD014,Meera Das,AIML,84,86,80,85,82,96%

16 STD015,Nikhil Sen,AIML,73,69,77,74,71,83%

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=====STUDENT DATA =====
Student_ID      Name Department Python DBMS DSA OS JavaScript Attendance
0 STD001 Aarav Sharma AIML 85 78 92 80 91 92%
1 STD002 Ananya Iyer AIML 94 88 85 65 74 96%
2 STD003 Rohan Verma AIML 62 58 70 80 91 90%
3 STD004 Sneha Patel AIML 78 82 75 72 55 74%
4 STD005 Vikram Rao AIML 88 91 84 86 89 88%
5 STD006 Priya Nair AIML 71 75 88 82 80 95%
6 STD007 Rahul Gupta AIML 55 64 60 58 70 81%
7 STD008 Pooja Kulkarni AIML 90 85 94 90 92 92%
8 STD009 Aditya Singh AIML 82 79 78 75 87 94%
9 STD010 Kavya Joshi AIML 67 72 81 79 84 89%
10 STD011 Manish Mehta AIML 76 70 68 66 73 85%
11 STD012 Divya Menon AIML 95 92 89 91 96 98%
12 STD013 Arjun Reddy AIML 58 62 72 70 60 76%
13 STD014 Meera Das AIML 84 86 80 85 82 96%
14 STD015 Nikhil Sen AIML 73 69 77 74 71 83%

Number of Students: 15
Columns: ['Student_ID', 'Name', 'Department', 'Python', 'DBMS', 'DSA', 'OS', 'JavaScript', 'Attendance']

===== STUDENT PERFORMANCE =====

Name: Aarav Sharma
Total: 426
Average: 85.2
Grade: A
Attendance: 92%
Result: Pass

Name: Ananya Iyer
Total: 406
Average: 81.2
Grade: A
Attendance: 96%
Result: Pass
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Name: Meera Das
Total: 417
Average: 83.4
Grade: A
Attendance: 96%
Result: Pass

Name: Nikhil Sen
Total: 364
Average: 72.8
Grade: B
Attendance: 83%
Result: Pass

===== SUBJECT-WISE AVERAGE =====
Python : 77.2
DBMS : 76.73
DSA : 79.53
OS : 76.87
JavaScript : 79.67

===== CLASS ANALYSIS =====
Students Passed: 15
Students Failed: 0
Highest Average: 92.6
Highest Average Student: Divya Menon
Lowest Average: 61.4
Lowest Average Student: Rahul Gupta
Class Average: 78.0
Best Subject: JavaScript

===== TOP PERFORMER =====
Name: Divya Menon
Average: 92.6
Grade: A+
```

8. Final Outcome

The project successfully analyzes student performance using Python, NumPy and Pandas. It provides individual student performance, subject-wise averages, pass/fail information, class performance and the top-performing student in a clear output format.

9. Challenges & Learning

Challenges:

- Setting up the Python environment and installing required libraries.
- Preparing the CSV dataset with the correct column names.
- Connecting the CSV data with the Python program.

Learning:

- Learned how to read CSV files using Pandas.

- Learned basic NumPy numerical operations.
- Practiced functions, loops and conditions in Python.
- Learned basic data analysis using Pandas and NumPy.
- Learned how to organize and upload a project to GitHub.

10. GitHub Repository

GitHub Repository: <https://github.com/KIRTI016/Student-Performance-Analytic-System.git>